

Techno-Economic and Environmental Analysis of a Grid-Tied Solar PV System for an Institution of Higher Learning Using PVSyst: Case of the University of Botswana

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ABSTRACT

The focus of this study is to design and simulate a grid tied solar PV system for a higher institution of learning using University of Botswana, Faculty of Engineering and Technology building as a case. The grid-connected system aimed at reducing the energy bill and contributes to the reduction of the CO₂ emissions. The system uses the physical data and PVSyst software and OpenSolar and google earth to determine the key parameters crucial for the design of the project. Technical performance analysis of the proposed rooftop PV plant showed that the energy injected into the grid, with consideration of all losses amounts to 480MWh/year with a substantial lifetime carbon reduction of 991.2tons. The economic evaluation of the designed system showed that the project is profitable at the discontinued payback period of 8.63 years and the annual energy bill reduction of 32% is achieved. These results are beneficial to institutions and any other entity that would like to migrate from their use of conventional power in that they will gain an insight on their CO₂ emission reduction and contribution to the national quota, their expected energy and cost savings. Governments and utilities can also benefit from these results in integrating with their energy and CO₂ emissions.

Keywords: PV System, Grid-tied, CO₂ Emissions, Economic Analysis, PVSyst, OpenSolar

I. INTRODUCTION

The global increase in demand for electrical energy has led to the need for increased generation capacity in many countries, including Botswana. Although

conventional energy sources provide most of the base load, the world is contending to abate their full use because of their negative effects such as greenhouse gas emissions, land, water, and air pollution. Botswana is one of the countries that heavily rely on

coal for its power production, accounting for over 75% of the country's generation capacity [1-3]. As the country's standard of living and industries evolve, power demand continues to grow and in addition energy prices increase. This puts consumers at a disadvantage. To address these, implementation of more sustainable ways to increase generation capacity, reduce pollution and costs for consumers has become inevitable.

Botswana has a significant solar potential receiving over 3,200 hours of sunshine per year with an average insolation on a flat surface of 21 MJ/m. This rate of irradiation is among the highest in the world [2-4]. Despite this, the solar industry in Botswana remains largely untapped. Since the early 1990, the country has been making some attempts to integrate solar in the energy mix but to date not much has been achieved. In an effort to increase participation in the solar energy sector, the country recently put into effect a net metering solar rooftop program. The program system-wide aggregate capacity of 10 MW for the first 12 months of the program was rolled out by BPC in 2020 [5,6]. The country has also amended its electricity supply act to allow Independent Power Producers (IPPs) participation in the electricity sector. This great development has set Botswana on a momentum to look into the integration of Renewable Energy Sources (RES) from small to large scale systems. Renewable energy capacity of more than 300MW has been projected for the planning period 2020-2040 [5-8]. It is envisaged that by 2030, 18% of Botswana's power will be generated from renewables [8].

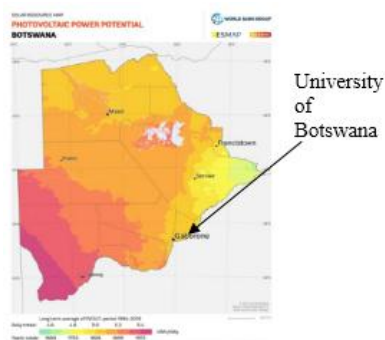


Fig 1: Photovoltaic Electricity potential-Botswana [4]

Botswana is a party to the United Nations Framework Convention on Climate Change (UNFCCC). Having adopted the Paris Agreement, in 2015 the government has made a voluntary commitment through its first Nationally Determined Contribution (NDC) to reduce the country's total greenhouse gas (GHG) emissions by 15% by the year 2030, with reference to 2010 emissions. Furthermore, the country is working towards providing affordable and clean energy to its general population in order to align with UN 2030 Agenda for Sustainable Development Goals, especially SDG7 and 13 [6-10]. Many local entities are now in a marathon to become the key contributors to these national goals. University of Botswana (UB) as one of the higher institutions of learning in Botswana has become one of such entities that have committed to contribute to this national pursuit [1].

Currently the University, other than having converted their borehole pumping systems to solar, is looking for cleaner methods of generating more power on campus with the aim of meeting a larger part of its day-time electricity needs to reduce the over BWP3.6 million (USD 266002.92) monthly bill. Further, these methods are anticipated to lessen the impact of global warming and create opportunities for the university to sell excess power to the national grid [11]. The institution has an immense roof top space that lay unutilized yet bear potential for the installation of PV systems. This work proposes a design of a PV system that can be installed in one of the roof tops and also provides an evaluation of the usefulness that this system can bring.

II. PROBLEM DESCRIPTION

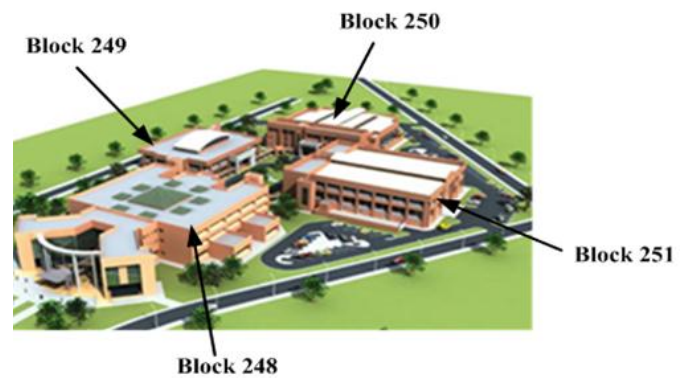
The University of Botswana (UB) is a tertiary institution in Botswana, Africa, posing as the main and oldest university in the country. It is located in the capital city of Botswana, Gaborone. It was established in 1964 and currently has 3 campuses (Gaborone, Francistown and Maun). The University offers 9 faculties with a total of 195 programs. It has a

total of 18,000 students and 2,658 staff members [13-15]. Of the three campuses, the study was performed on main and largest campus, Gaborone. UB has significant and state of the art facilities as shown in Figure 2.



Figure 2: Some of the UB Main Campus Infrastructure

This study was conducted in the Faculty of Engineering & Technology (FET). The faculty consists of the magnificent state of the art facilities (Blocks 248-251) as shown in Figure 3. FET consists of 5 departments; architecture & planning, civil, electrical, Industrial design and, Mechanical & industrial engineering. The faculty, on average, accommodates 1,600 students every year [16] with a staff complement of 133 (academic & support). This study was performed in the largest block 248.



(a)



(b)

Fig 3: (a) UB Faculty of Engineering & Technology, (b) View of block 248

Block 248 houses a total of 120 staff offices (academics and support), twelve (12) class rooms and six (6) computer labs. Daily average energy usage during the school term is around 1,394.5kWh. The daily average electricity cost for the building is P1,087.68 (USD80.38).

The main aim of this work is to design and simulate a solar PV system for the University of Botswana, Block 248 to generate electricity that will supply the building with the possibility to produce surplus to the grid and estimate the CO₂ emissions.

III.DATA

This section presents the data collection process that aided the determination of energy consumption. The methodology used estimates the energy consumption of all electrical appliances in the building. This was done by calculating the power consumption of each appliance from power ratings and the estimated

number of hours committed per day. The summary of the data is presented in Table 1.

Table 1: Summary of the loads and their energy consumption

Type of load	Total Energy consumption (kWh)
Lighting	5,763.3
Computer devices	652.9
Surveillance	59.5
HVAC	537.60
Projectors	3.6
Water fountains	241.9
Printer/Photocopiers	39.6
Fire devices	16.6
Elevators	2,490.00
Total energy consumption	9,761.70

From Table 1, the total weekly approximate energy consumption of the building is 9.76MWh and using the Botswana Power Corporation (BPC's) commercial tariff, this would attract a weekly bill of BWP7,613.58 (USD 561.30). The average day consumption would be 1,952.34MWh with a bill of BWP1, 522.72 (USD 113.11). This total energy consumption was used in the subsequent steps for designing a solar PV system bearing in mind that availability of excess power will be used to feed other sections on campus or fed to the grid.

The computation to determine solar panels needed is then given as follows:

Taking the Peak Sun Hours for Gaborone (University of Botswana) as 6hours as obtained from Atmospheric Science Data Center (NASA) online database. The plant size in KW with an efficiency factor of 15%;

$$\frac{\text{Daily energy consumption (kWh)}}{\text{Sun Peak Hours (h)}} * \eta \text{ factor} \\ = \frac{1952.34 \text{ (kWh)}}{6 \text{ (h)}} * 1.15 = 374.2 \text{ kW}$$

A Monocrystalline with a panel rating ranging from 565Wp is used for the proposed project. Then,

$$\frac{\text{Total Design wattage (W)}}{\text{individual panel watts (w)}} = \frac{374200 \text{ (W)}}{565 \text{ (W)}} \\ = 662 \text{ solar panels}$$

The rooftop of block 248 has a flat roof area of 2695.81m² and the view is provided in Figure 3. Table 2 provides additional vital information about the location.

Table 2: Site Information

Site name	Block 248 Building (University of Botswana)
Site coordinates	24°39'43"S 25°56'05"E
Area	2695.81m ²
Slope inclination	0°
Slope azimuth	-129°
Annual global in-plane irradiation	$\frac{6640 \text{ Wh}}{\text{m}^2} / \text{day}$

IV. RESULTS AND DISCUSSION

Table 3 presents the characteristics of the system PV components and their respective ratings. The Jinko solar PV module is well known for delivering reliable, economically efficient solar panels tailored to both commercial and residential. The solar panel has a higher nominal power and this is advantageous because it typically has advanced technology which translates to greater efficiency in converting sunlight into electricity [17]. Efficiency is crucial for maximizing energy production, especially during suboptimal conditions. For the given rooftop area, 512 PV modules can be used arranged in 32 strings *16 in series. The variation to the theoretical estimate is justifiable as the entire area is not usable, sunroof and also allowance for frames, cabling and other accessories.

Table 3 and 4 present the design parameters for the PV modules and the inverter. The total area occupied

by panels only is 1400m² with gross panel weight of 15, 924kg. The modules are laid out as shown in Figure 3 on the concrete roof which is regarded as the one of the best material that can withstand such load [18, 19]

The rooftop panel layout was drawn using OPENSOLAR that efficiently incorporated data obtained from the PVSyst such as the string number information. A string in this system has 16 panels connected in series. It is obtained that inverters will have 16 panels by 7 strings, while the three other inverters will have 16 panels by 6 strings.

Table 3: Design Parameters of the PV module

Design Parameters	Design Quantities
PV Module Model	JKM565-7RL4-V
	Mono-facial
Unit Nominal Power	565Wp
No. Of PV Modules	512
Nominal (STC)	289kWp
Module arrangement	12 strings *16 in series
Maximum Power Point (MPP) parameters at 50°C	
P _{mpp}	264kWp
V _{mpp}	638V
I _{mpp}	414A
Other Important Parameters	
Cell type	P type Mono-crystalline
Module Dimensions	2411×1134×35mm
Module Efficiency	20.67%
Nominal operating cell temperature (NOCT)	45+-2°C
Weight	31.1kg

Table 4: Design Parameters of the Inverter

Design Parameters	Design Quantities
Inverter Module	Tauro Eco 50-3-D-208-220
Model	+30A fuse option
Unit	Nominal 50.0kWac
Power	
No. Of inverters	5
Total power	250kWac
Operating voltage	500-850V
P _{nom} ration (DC-AC)	1.16
DC input voltage range	U _{dc} min - U _{dc} max: 580 - 1000 V
AC output current	I _{ac} nom:, 72,5A at 230Vac

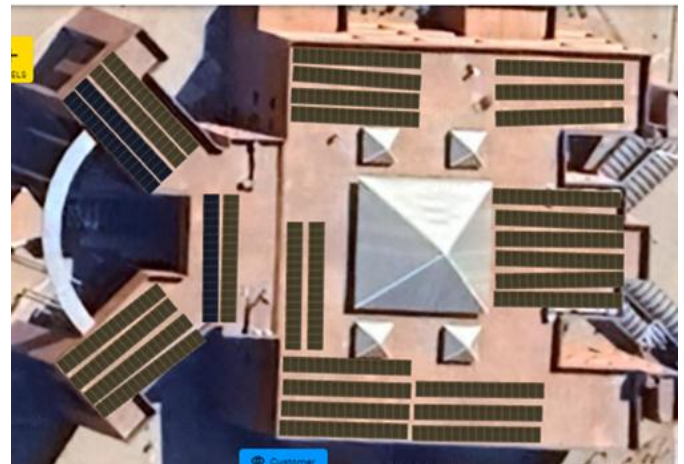
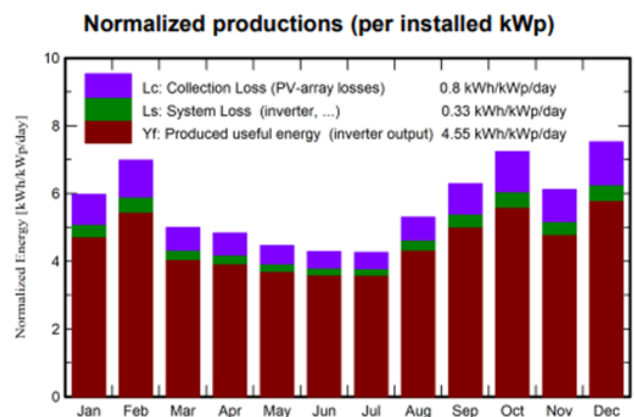


Figure 4: Panels layout.



(a)

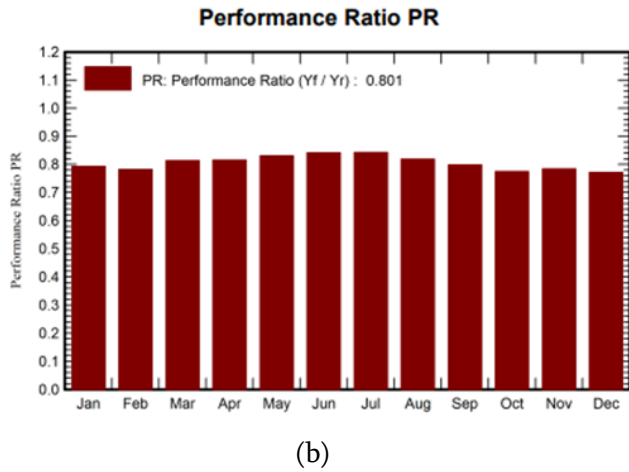


Figure 5: a) Normalised energy production per month.
b) Performance ratio (PR)

Fig 5(a), presents the total annual energy produced by the system, 480,364 kWh. The results represent the peak months for the energy production, as well as months with the lowest energy production. The results suggest lower production during winter months of May-July. Summer months record high energy production. In Botswana, Gaborone has mean monthly maximum temperatures ranging from 29.5°C to 35°C summer, and 19.8°C to 28.9°C in winter [20]. The performance ratio (PR) of the designed system has an average of 80% which is within the allowable limits of a good system [21]. A system with a PR of less than 60% is deemed to be inefficient whilst more than 95% is classified as too perfect and impractical [22]. According to Fig 4b, the system performs at its maximum during winter. The best possible explanation for this is that cold sunny weather improves the efficiency of the solar panels [2, 21- 23]. The lowest performance recorded is 77.1% which is in December a summer month, proves that indeed solar cells work more efficiently in winter [23,24]. Figure 5 provides the monthly average energy balance and PR of the system

	GlobHor kWh/m ²	DiffHor kWh/m ²	T_Amb °C	GlobInc kWh/m ²	GlobEff kWh/m ²	EArray kWh	E_Grid kWh	PR ratio
January	185.1	79.08	23.93	185.1	181.3	45707	42436	0.793
February	195.6	60.29	24.53	195.6	191.7	47830	44238	0.782
March	154.9	68.36	20.79	154.9	150.7	38937	36420	0.813
April	144.9	49.29	20.24	144.9	140.5	36406	34159	0.815
May	138.5	34.71	16.81	138.5	132.9	35250	33274	0.831
June	128.6	28.65	13.42	128.6	122.8	33059	31291	0.841
July	132.1	33.16	13.68	132.1	126.3	34024	32189	0.842
August	164.3	38.87	16.91	164.3	158.9	41579	38923	0.819
September	188.5	49.26	20.84	188.5	184.0	46880	43547	0.799
October	224.4	57.31	25.24	224.4	220.2	54373	50257	0.774
November	183.5	69.09	23.55	183.5	179.7	44945	41646	0.784
December	233.1	72.42	26.16	233.1	228.7	56231	51985	0.771
Year	2073.4	640.47	20.49	2073.4	2017.7	515221	480364	0.801

Figure 5: Monthly average energy balance and performance ratio (PR).

The system comprehensive losses are presented in Figure 6.

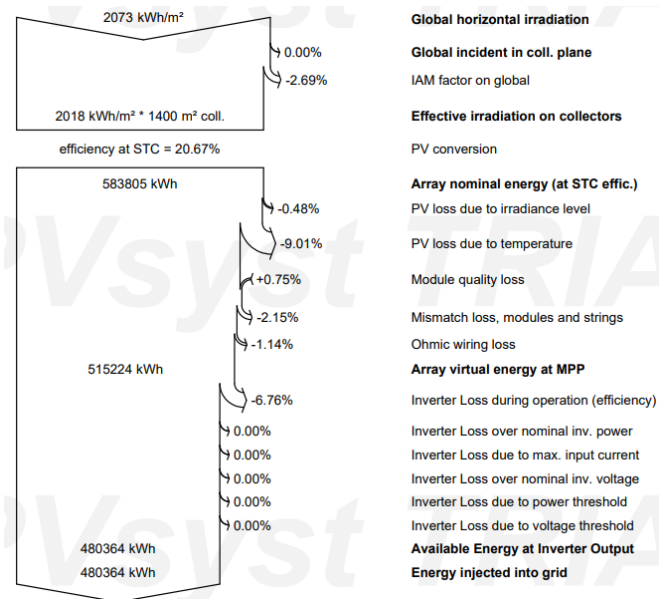


Figure 6: System loss diagram.

V. ECONOMIC ANALYSIS

There are different methods used to perform economic analysis of systems in the literature. These methods include but not limited to net present value (NPV), pay back period (PB) and discounted present value (DPV) [25-28]. In this paper we adopt the DPV method to calculate the length of time to recoup an investment on usage of PV system based on the investment's discounted cash flows. By discounting each individual cash flow, the discounted payback period takes into consideration the time value of money.

$$DPV = \frac{FV}{(1 + r)^n}$$

DPV; Discounted present value, FV; Future Value, r ; Rate of Return, n ; number of periods.

r is as 6% for the current month, May [29]

In order to make economic analysis of the PV system, certain assumptions are made.

The PV costs entail capital cost, operational and maintenance (O&M) costs as shown in Table 6. Since our study horizon is one day we annualize our costs.

Table 6: Approximate cost of components

Component	Amount (BWP)
Solar panels	1,708,188.93
Inverter, Protection & accessories	625,919.71
Mounting frames, cabling & other accessories, Installation cost	429,566.71
<i>Sub-Total</i>	<i>2,763,675.35</i>
O&M at 2.5%	6,9091.88
Total	2,832,767.234

The O&M cost of a PV has been estimated to a specific annual value or some online sources estimate the annual O&M cost to be around 2-2.5% of capital cost and in this work we use less conservative value of 2.5%. Calculation of savings brought by the use of the PV system is performed as follows. The annual cost of energy consumed from the baseline data, CAEC;

$$CAEC = \frac{E_c}{day} * 365 * D_f * \xi$$

$$= BWP 354,590.76 kWh/yr$$

$\xi = \rho t$ is the cost of energy. For the annual consumption we assume a flat two part energy tariff for government derived from the BPC tariff structure [30].

The annual cost of energy from the PV is given;

$$CAEO = \sum_{t=1}^T P_{pv} * \Delta t * \frac{365}{12} * \xi$$

$$= BWP 468,262.2 kWh/yr$$

The percentage annual energy saving, %AES, that is brought by the use of the PV system

$$\%AES = \frac{AEO - AEC}{AEO} * 100$$

$$= \frac{502,020.00 - 379,025.28}{502,020.00} * 100$$

$$= 24.50\%$$

The annualised cost savings (AES) due to the PV system is determined from the product of the annual bill charge and %S.

Table 7: Energy savings & Cost savings due to the PV system

AEO (KWh/yr)	AEC (kWh/yr)	%AES (%)	% (%)	CAES
1,255,050	379,025.28	24.50	32.06	

The results in table 7 show that the use of PV system yields an energy cost saving of 32.06%. This is within reasonable values of 20-45% reported in most literature. The earning is used as the optimal benefit of using PV system in calculating the discounted present value.

Table 8: Payback Period

Years	0	1	2	3	4	5	6	7	8	9
Solar photovoltaics	(1 708 188.93)									
Inverters and accessories	(625 919.71)									
Mounting frames, cabling & other accessories, Installation cost	(429 566.71)									
Sub total (Capital cost)	(2 763 675.35)									
Operation & maintenance cost @ 2.5% (Capital cost)		(69 091.88)	(69 091.88)	(69 091.88)	(69 091.88)	(69 091.88)	(69 091.88)	(69 091.88)	(69 091.88)	(69 091.88)
Saved costs due to PV		468 262.2	468 262.2	468 262.2	468 262.2	468 262.2	468 262.2	468 262.2	468 262.2	468 262.2
Discount factor @ 6%		0.94	0.89	0.84	0.79	0.75	0.70	0.67	0.63	0.59
Discounted cashflows	(2 763 675.35)	376 575.77	355 269.36	335 151.09	316 189.28	298 283.38	281 399.32	265 471.06	250 444.29	236 268.30
Discounted Payback Period	Years	Discounted cashflows	Cumulative cashflows							
	0	(2 763 675.35)	(2 763 675.35)							
	1	376 575.77	(2 387 099.58)							
	2	355 269.36	(2 031 830.22)							
	3	335 151.09	(1 696 680.32)							
	4	316 189.28	(1 380 491.05)							
	5	298 283.38	(1 082 207.67)							
	6	281 399.32	(800 808.34)							
	7	265 471.06	(535 337.28)							
	8	250 444.29	(284 892.99)							
	9	236 268.30	(48 624.69)							
Payback is:	8.63 years									

Table 8 presents the payback period and the assumptions made are that the O&M costs and the benefits are constant throughout the projected years into the future. This assumption implies that exclusion of weighted sum of capital cost (Wacc) the pay-back period may be reduced. It can be seen from the table that in this case, the payback period of 8.63 years is realised. A more precise results could be obtained from

the actual sizing of the PV system and consideration of the Wacc.

VI.CO₂EMISSIONS

The CO₂ emission factor for Botswana, 1.78 kg/kWh [31,32], therefore;

The total net mass of the CO₂ emission in kilogram for the lifetime of the solar PV system is;

$$M_t = \sum_{t=1}^N (E_{consumption} - E_{pv}) * \alpha_{grid} [33]$$

$$= (507,624.0 - 480,3664.00) \times 20 \text{ years} * 1.78 = 899,256 \text{ kg or } 991.2 \text{ ton of CO}_2$$

VII.CONCLUSION

The proposed system design includes 512 solar panels, which will be installed on the rooftop of block 248. The panels will be connected to 5 inverters, which will convert the DC power generated by the panels into AC power that can be used by the facility. Technical performance analysis of the proposed rooftop PV plant results showed that the energy injected into the grid, with consideration of all losses amounts to 480MWh/year with a substantial carbon reduction of 991.2 tons. The economic evaluation of the proposed PV plant was also performed in this study and results showed that the project could potentially recoup its cost in 8years and an energy cost saving of 32 %. These results are beneficial to institutions and any other entity that would like to migrate from their use of conventional power their CO₂ emission reduction contribution to the national quota.

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